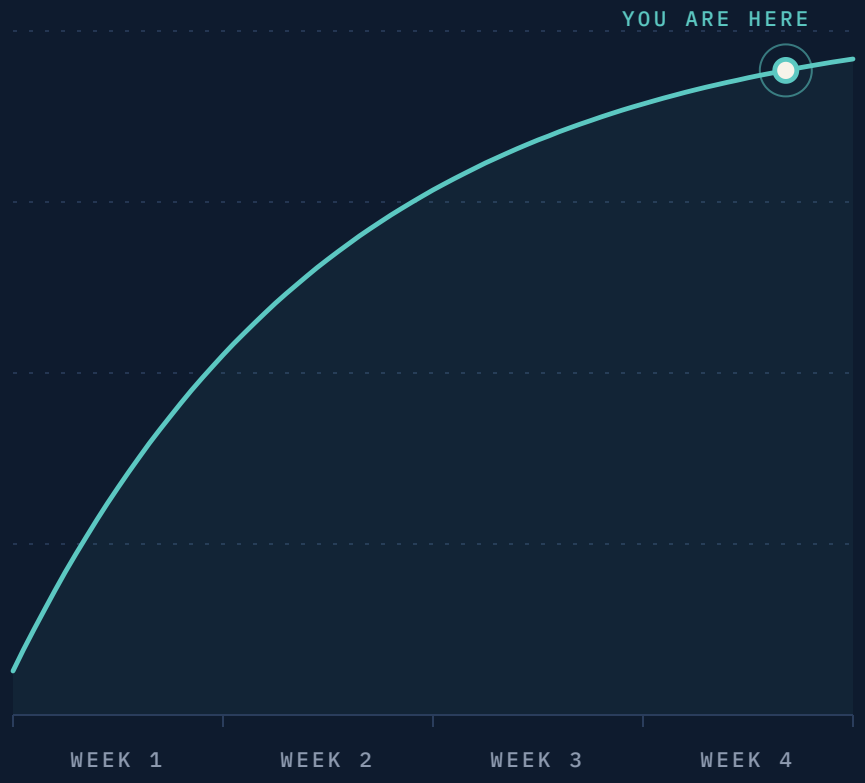
WEEK 4 · DAY 3 · OVERVIEW

Case Study

Statement Sentiment



Created by **Jeff Lenamon**

ABOUT 55 MIN WITH EXERCISES / 7 SECTIONS / 7 EXERCISES

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Problem Definition

Analyzing Text Part 2 gave us clean tokens for all 46 FOMC statements. Give me one number per statement for its wording, three ways: two lists we type, a published list, and a list built from the statements' own words. Show me the words behind every number, and whether the three agree. Last: does a change in a score line up with that day's 2 Yr and 10 Yr moves? Describe only: no reading of policy, no view on rates.

Your Route Through Day 3

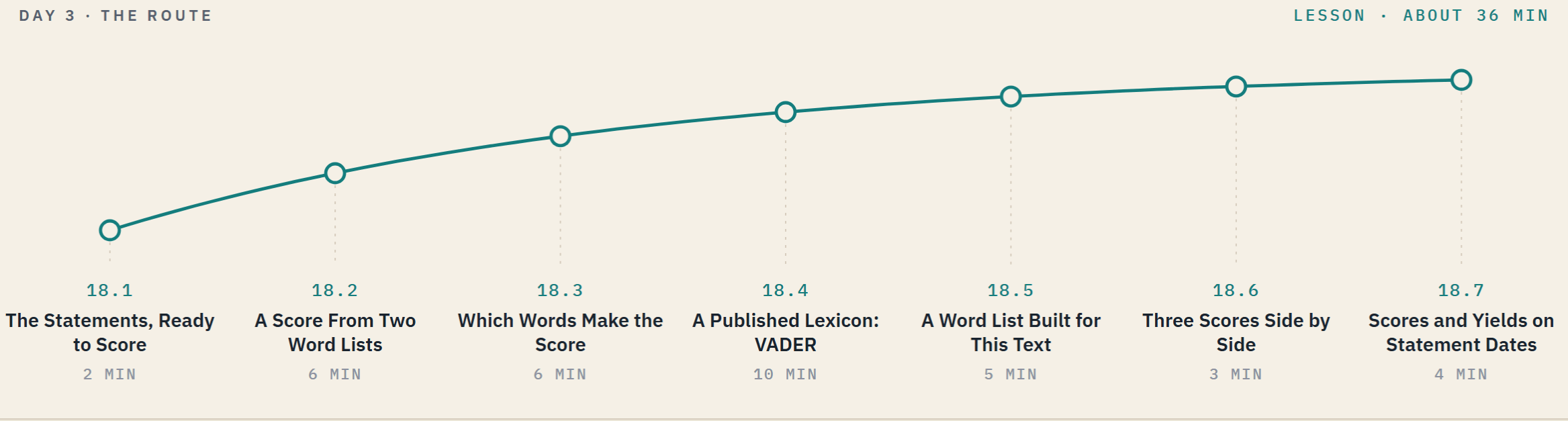


FIGURE 2 Seven sections, in order, with about how long each takes.

Time. About 36 minutes for the lesson, then about 21 for the seven exercises.

18.1 The Statements, Ready to Score

LOAD THE STATEMENTS AND CHECK THEM

```
statements = pd.read_csv(data_folder + 'fomc_statements.csv')
statements['meeting'] = pd.to_datetime(statements['date'], format='%Y-%m-%d')
print(statements.shape, statements.isna().sum().sum(), statements.duplicated().sum())
```

```
(46, 4) 0 0
```

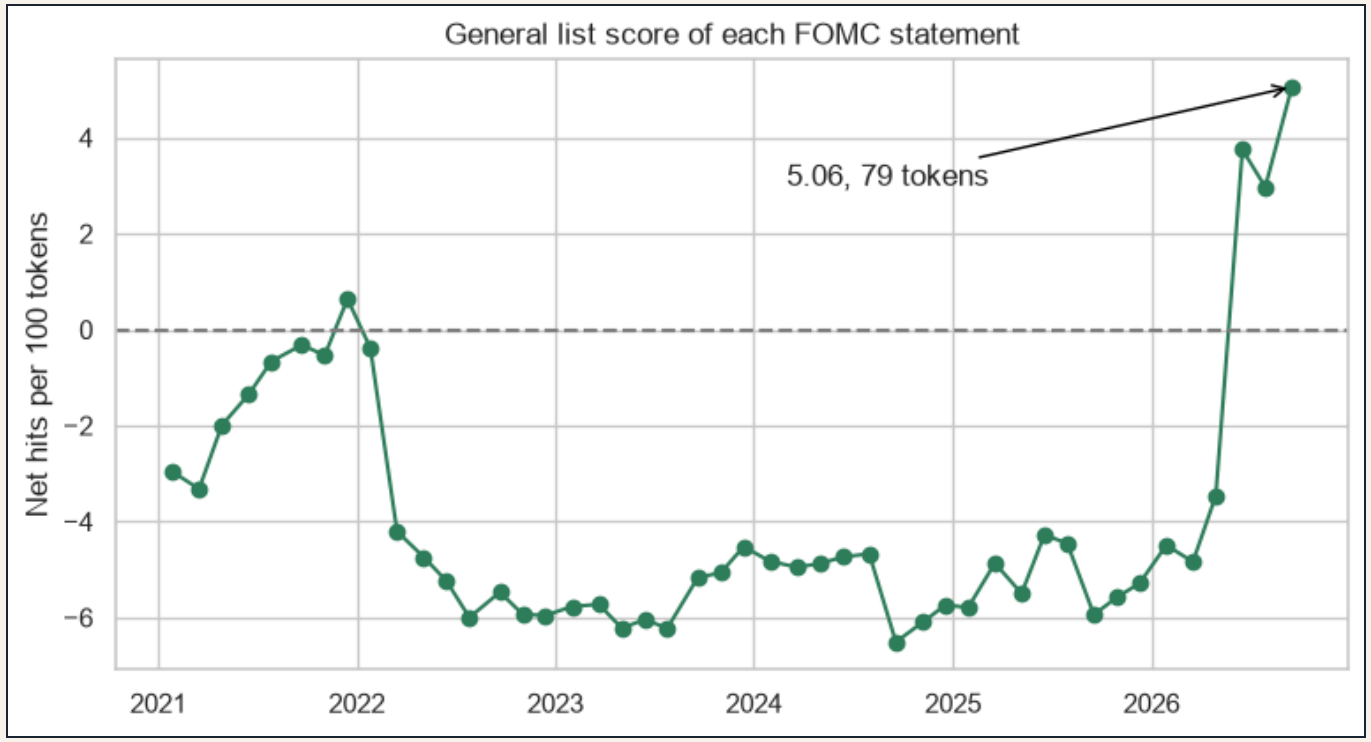
CLEAN EVERY STATEMENT, DROP THE STOPWORDS, COUNT THE TOKENS

```
print(statements['n_kept'].sum(), statements['n_kept'].median(), statements['n_kept'].max())
print(statements.nsmallest(4, 'n_kept')[['date', 'n_kept']])
```

```
10204 214.0 372
         date  n_kept
45  2026-09-16      79
```

The three shortest, 79 to 101 tokens, are the last three statements. Keep that in mind.

18.2 A Score From Two Word Lists



SCORE ALL 46, WHAT IT PRINTS

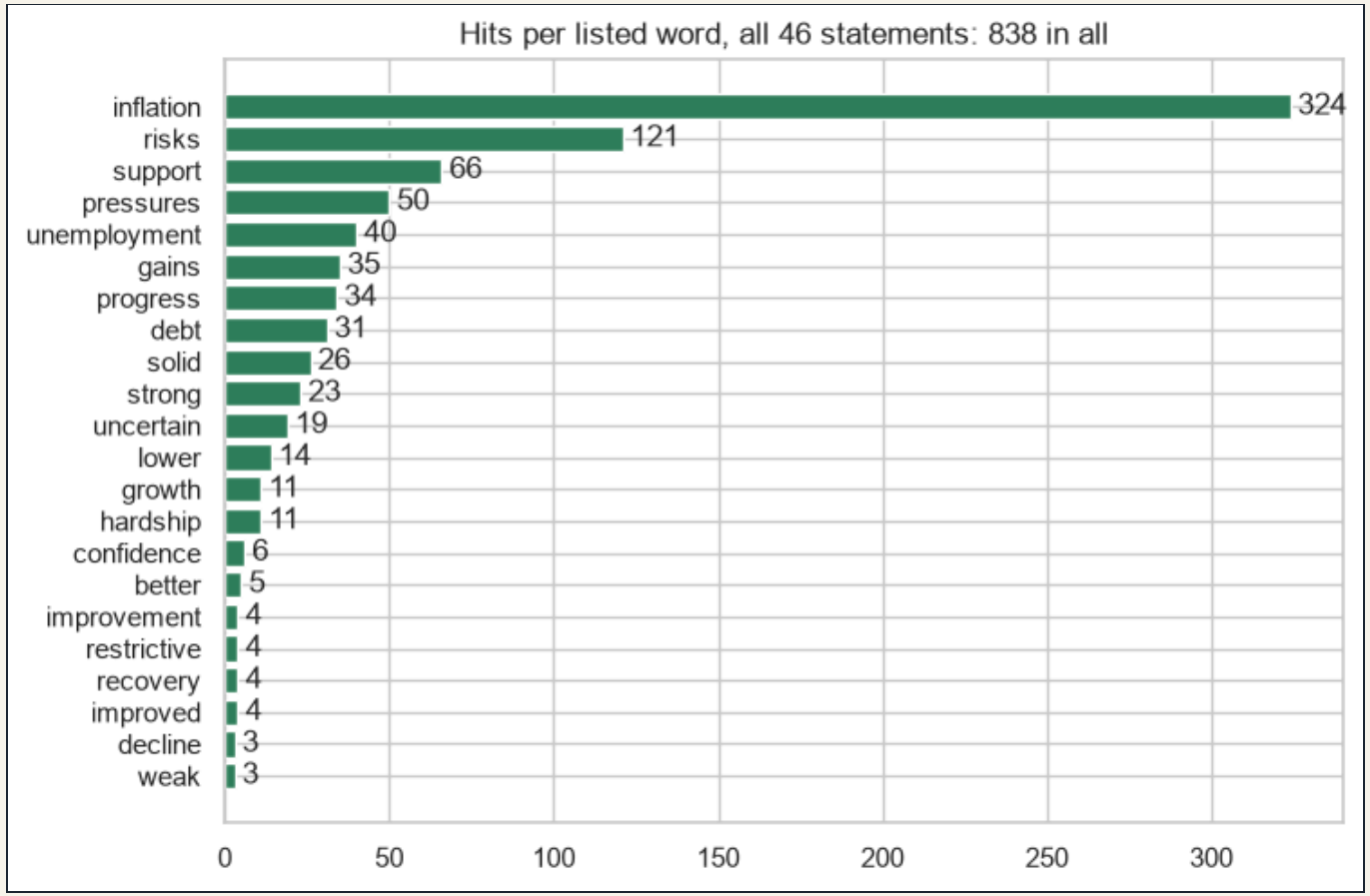
```
2024-09-18 -6.51
2026-09-16 5.06
42 -3.88
```

PIECES OF TEXT, WHAT IT PRINTS

```
6
17
4
```

42 of 46 below zero. Before trusting it, look at the words.

18.3 Which Words Make the Score



COUNT THE HITS PER WORD

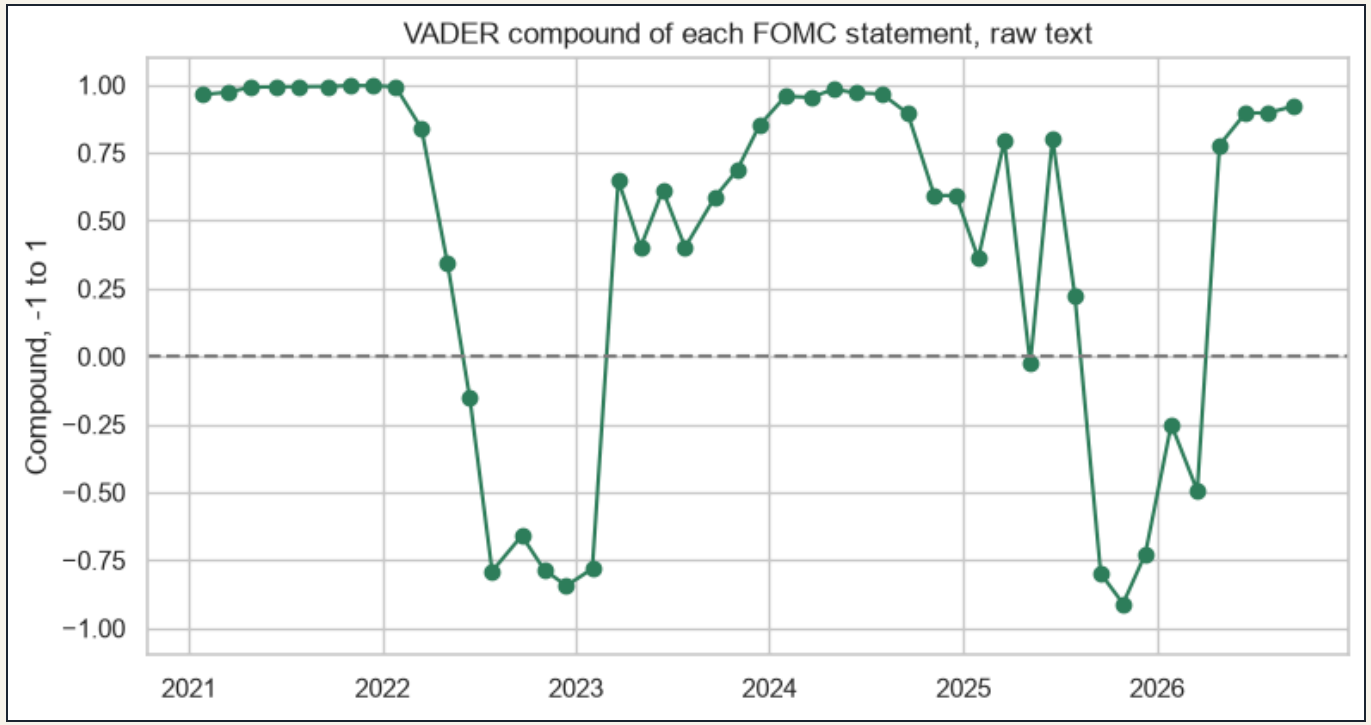
```
hits = Counter()
per_word = pd.Series(hits).sort_values()
```

THE SEVEN TERMS OF THE TRADE

```
628 of 838
```

`inflation` **makes 324 of the 838 hits, more than every positive word together;** good **and** bad **never hit, so they have no bar.**

18.4 A Published Lexicon: VADER



LOOK UP WORDS

```
print(len(analyzer.lexicon))
for word in look_up:
```

```
7502
good 1.9
bad -2.5
strong 2.3
weak -1.9
inflation absent
securities 1.2
treasury 0.8
```

Of 7,502 rated words, `inflation` is not one: the word behind 324 hits counts for nothing.

The Cut-Off Moves the Counts

LABELS AT FIVE CUT-OFFS

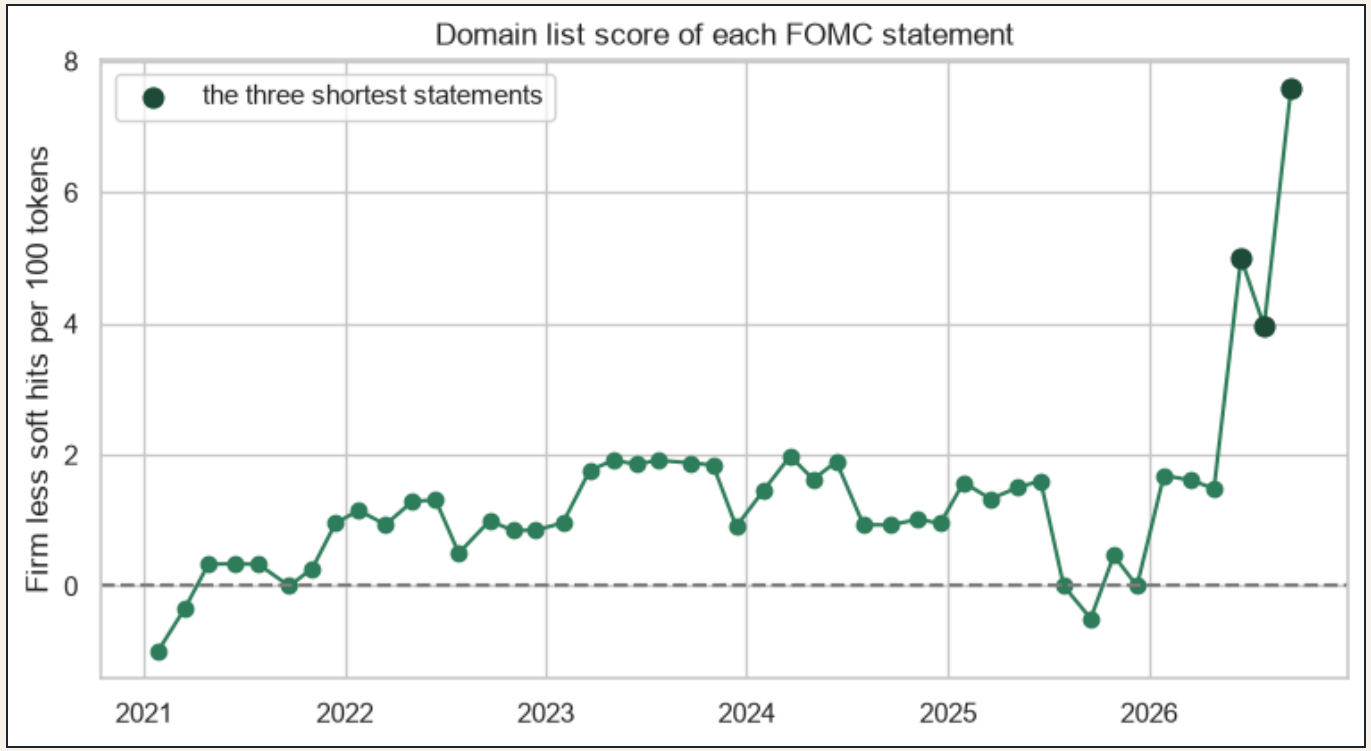
VADER	NEGATIVE	NEUTRAL	POSITIVE
0.05	11	1	34
0.25	10	3	33
0.5	8	9	29
0.75	6	17	23
0.9	1	30	15

EXCEL SIDE BY SIDE

JOB	PYTHON	EXCEL
Look up a rating	`.lexicon.get('strong', 0)`	`=XLOOKUP("strong", lexicon!A:A, lexicon!B:B, 0)` returns 2.3, and 0 for a missing word
The label	`label_score(compound)`	`=IF(E2>=$H$1,"positive",IF(E2<=-$H$1,"negative","neutral"))`, the cut-off in H1
Labels at a cut-off	`value_counts()`	`=COUNTIF($E$2:$E$47,">="&H2)` and `"<="&-H2` return 29 and 8 at 0.5

The cut-off moves the counts as much as the text does. A list built for this text is next.

18.5 A Word List Built for This Text



COUNT EVERY TOKEN

```
all_counts = Counter()
```

```
507 [0, 0, 0]
133
26
```

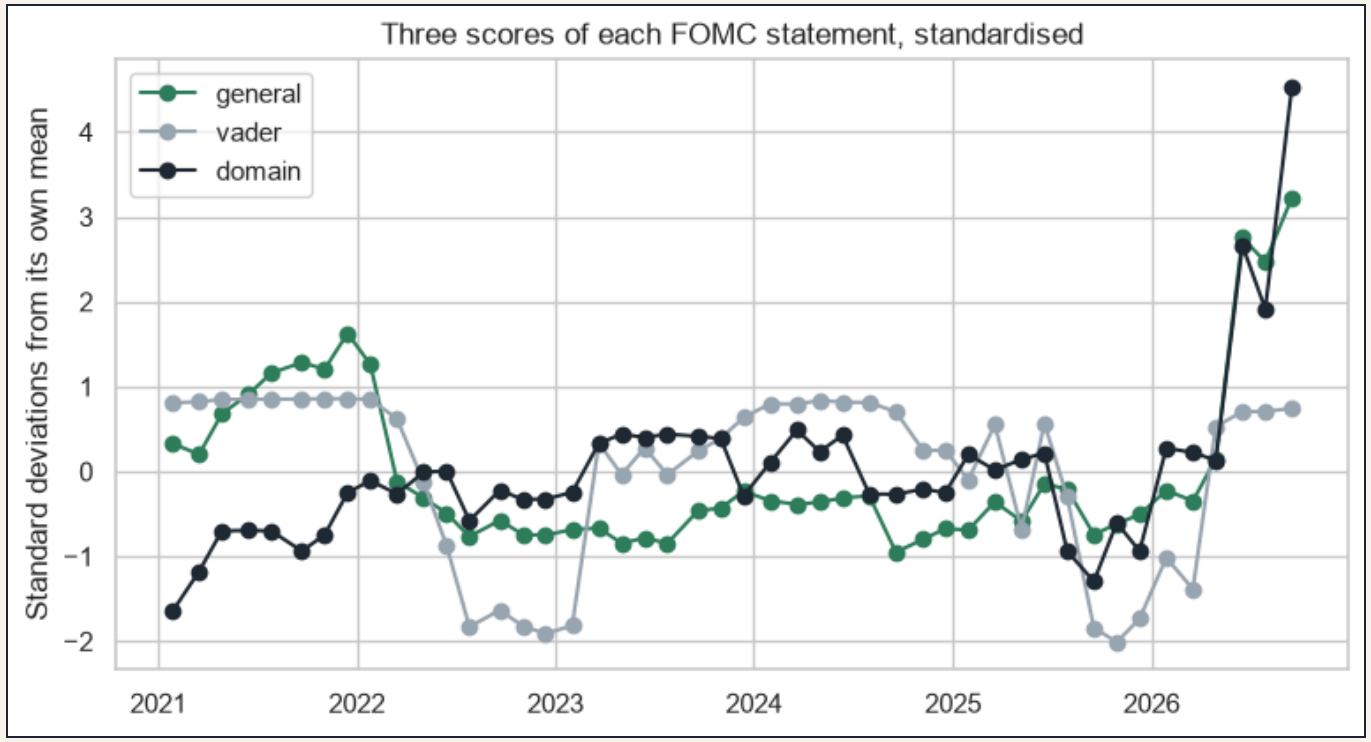
SCORE WITH THE BUILT LIST

```
def domain_score(tokens):
```

```
1.3 3
[1.49, 5.0, 3.96, 7.59] [202, 80, 101, 79]
```

Built from the file, the list is still blind to which word describes which.

18.6 Three Scores Side by Side



HOW CLOSELY THEY MOVE

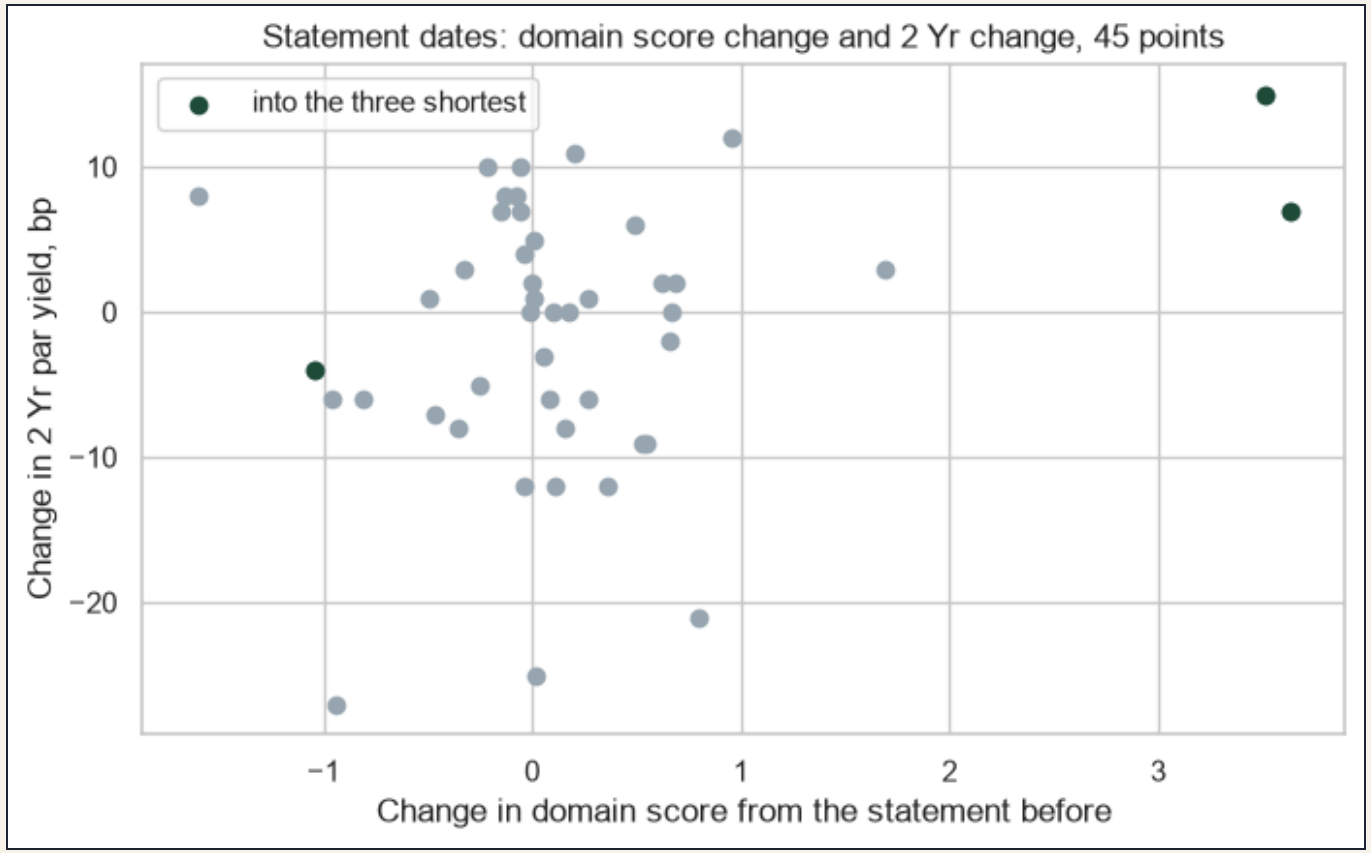
	GENERAL	VADER	DOMAIN
general	1.00	0.48	0.51
vader	0.48	1.00	0.21
domain	0.51	0.21	1.00

ONE SCALE

```
z = (scores - scores.mean()) / scores.std()
```

Three lists, one text, three answers. Last: yields.

18.7 Scores and Yields on Statement Dates



SIX CORRELATIONS ON 45 POINTS

	2 YR	10 YR
general	0.19	0.07
vader	-0.06	-0.03
domain	0.27	0.16

WITHOUT THE LAST THREE

```
without_last = joined.iloc[:-3]
```

```
          2 Yr  10 Yr
general  -0.07  -0.11
vader    -0.07  -0.04
domain    0.07   0.10
```

In this file, near zero: -0.06 to 0.27 on 45 points, -0.11 to 0.10 on 42.

You Can Now

- Score tokens against word lists, and print the words that hit.
- Read the sentences behind a number, with their dates.
- Use VADER on raw text, and list the words it rated.
- Label scores at a stated cut-off, and draw them as percent bars.
- Report a correlation with its number of points.

THE NOTE ASKED	ANSWER
One number, three ways	42, 12 and 3 of 46 below zero
The words behind them	**`inflation`** is 324 of 838 general hits
Do they agree	loosely: 0.48, 0.51, 0.21
Against yields	near zero, on 45 points and on 42

NEXT
WEEK 4 · DAY 4

From Notebook to Module

The EDA notebook's cleaning routine moves into a file of functions that any notebook can import.